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A mathematical model of obesity

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Remark 1. *The TITLE and ABSTRACT is Mandatory. moreover, if the manuscript is written in Spanish add RESUMEN o in Portuguese add RESUMO.*

Abstract

In this work, the work is investigated and presented precisely, indicating the new contribution of knowledge, the latter is important, because if it is not, it runs the risk of not being accepted.

Keywords. Ordinary differential equation, multilevel, Stability, Simulation.

1. Introduction. In the universal literature on mathematical modeling of obesity, there are two aspects of its study, one from the epidemiological point of view, since due to poor eating habits or physical activity, these have been propagation factors in the diverse populations of this chronic disease, obesity (population-based model) as in [1], [2] and the other point of view is physiological in terms of the type of food consumed (individual-based model) [3][4].

2. Model. For the construction of this model, we will consider three subpopulations of the total population, individuals with normal weight $N(t)$, individuals with overweight $S(t)$ and those individuals who are obese $O(t)$. Let's also consider some assumptions:

The transitions between these three subpopulations occur as follows:

- A person of normal weight $N(t)$ who begins to overindulge in unhealthy food begins his transition to an overweight person $S(t)$.
- An overweight person S(t) who continues with his excess of unhealthy food, becomes a person with obesity O(t).
- ...
- ...

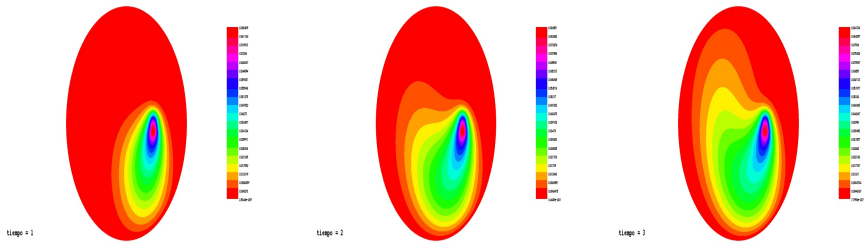
With all these considerations, the model is (see in [5])

$$\begin{aligned}
\frac{dN(t)}{dt} &= (1-\rho)\mu P(t) - \alpha\frac{S(t)}{P(t)}N(t) - \beta\frac{O(t)}{P(t)}N(t) - \mu N(t) \\
\frac{dS(t)}{dt} &= \rho\mu P(t) + \alpha\frac{S(t)}{P(t)}N(t) - (\epsilon+\mu)S(t) \\
\frac{dO(t)}{dt} &= \beta\frac{O(t)}{P(t)}N(t) + \epsilon S(t) - \mu O(t)
\end{aligned} \tag{2.1}$$

where the variables for the basic model are given in Table 2.1

Variables	Meaning
$N(t)$	Non-obese population at any time t (people with IMC less than 24.9)
$S(t)$	Overweight non-obese population at any time t (people starting to eat calories, i.e. with IMC between 25 and 29.9)
$O(t)$	Obese population at any time t (people with IMC greater than 30)
$P(t)$	Total population at any time t, $P(t) = N(t) + S(t) + O(t)$

Table 2.1: System Variable Meanings (2.1)



(a) Primera imagen. (b) Primera imagen. (c) Segunda imagen.

Figure 2.1: Dos imágenes en la misma figura.

2.1. System Reduction. When using sufigures, we have to use the following format
Consequently, the system (2.1) is reduced to the study of a Normalized two-dimensional system,

$$\begin{aligned} \frac{ds}{dt} &= \rho\mu + \alpha s[1-(s+o)] - (\epsilon+\mu)s \\ \frac{do}{dt} &= \beta o[1-(s+o)] + \epsilon s - \mu o \end{aligned} \tag{2.2}$$

2.2. Model Analysis. Existence of solutions;

Theorem 2.1. *For all $s_0, o_0 > 0$, exist $s, o : [0,\infty) \longrightarrow [0,\infty)$ that solve the system (2.2) with initial conditions $s = s_0, o = o_0$.*

Also. All solutions of the system (2.2) are bounded and positive.

Proof: The existence of the solutions is guaranteed by Peano's theorem. ... □

2.3. Simulations. Here is a simulation of the obesity model given by (2.1), with initial conditions dating from the year 2005 $(t = 0)$.

It can be seen from the simulation of the model (2.1), that in a non-obese population of 100, 35 of those who are overweight and 25 obese, if the conversion rates to the subpopulations at risk are small , we will have control over obesity. Our purpose will be to demonstrate this situation theoretically.

3. Conclusiones. In this work, we analyze obesity from an epidemiological point of view. The first describes the propagation of two states of abnormality in weight of individuals, without considering any treatment.

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acknowledgment.

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[1] Dawes J, González-Parra G, Rowley J. Enhancing the customer experience: contributions from information technology, J Business Res. 2005; 36(5):350-7.

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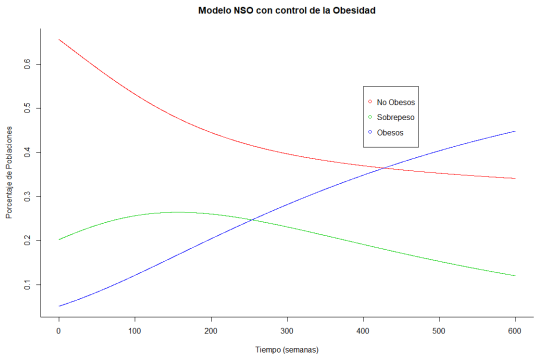

Figure 2.2: Uncontrolled model of obesity (2.1)

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Remark .1.

Proof: body of the environment proof finish with □

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